Module 13

Planning and configuring identify federation

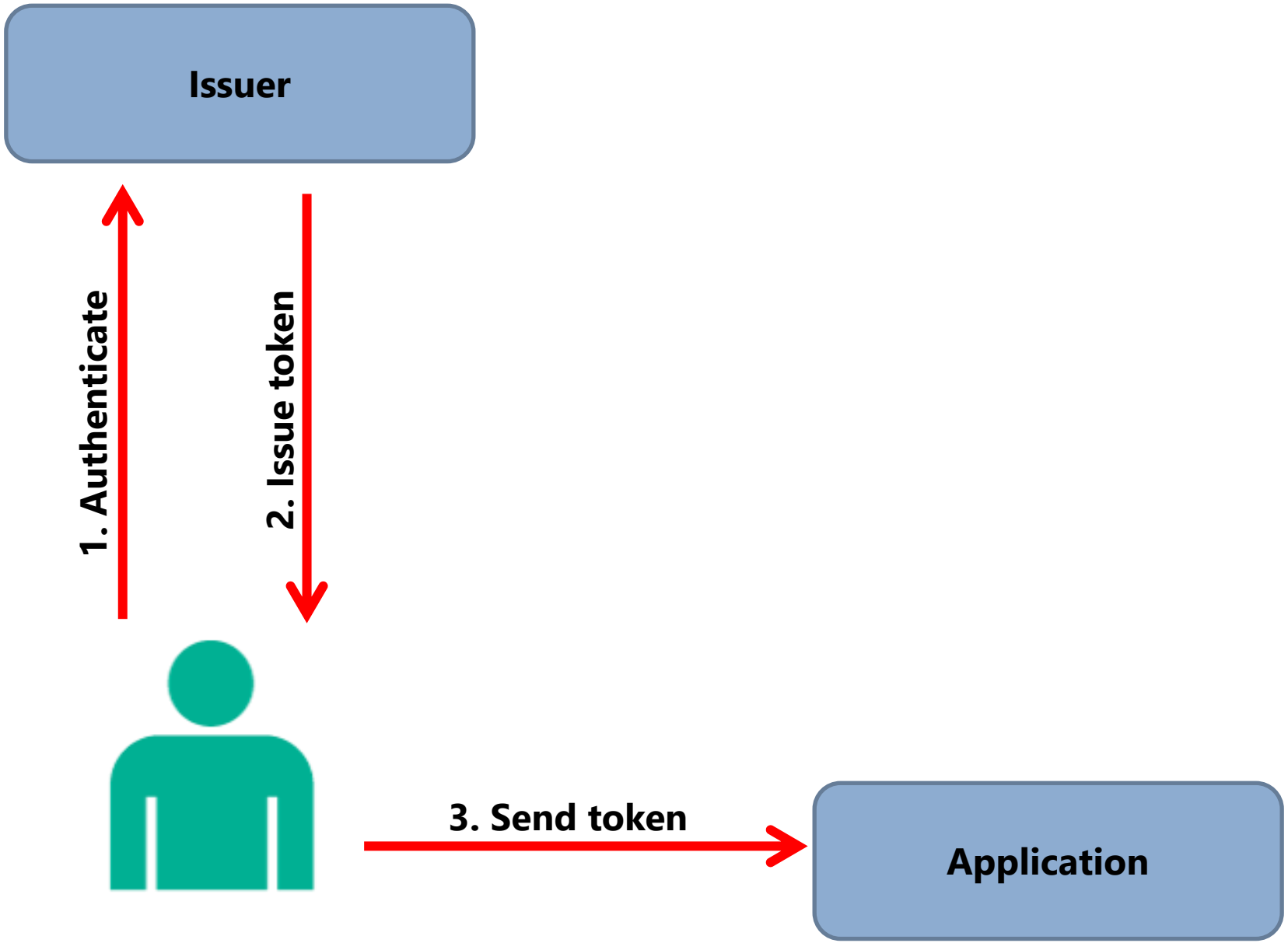
Module Overview

- Understanding identity federation
- Planning an AD FS deployment
- Deploy AD FS for identity federation with Office 365
- Planning and implementing hybrid solutions (Optional)

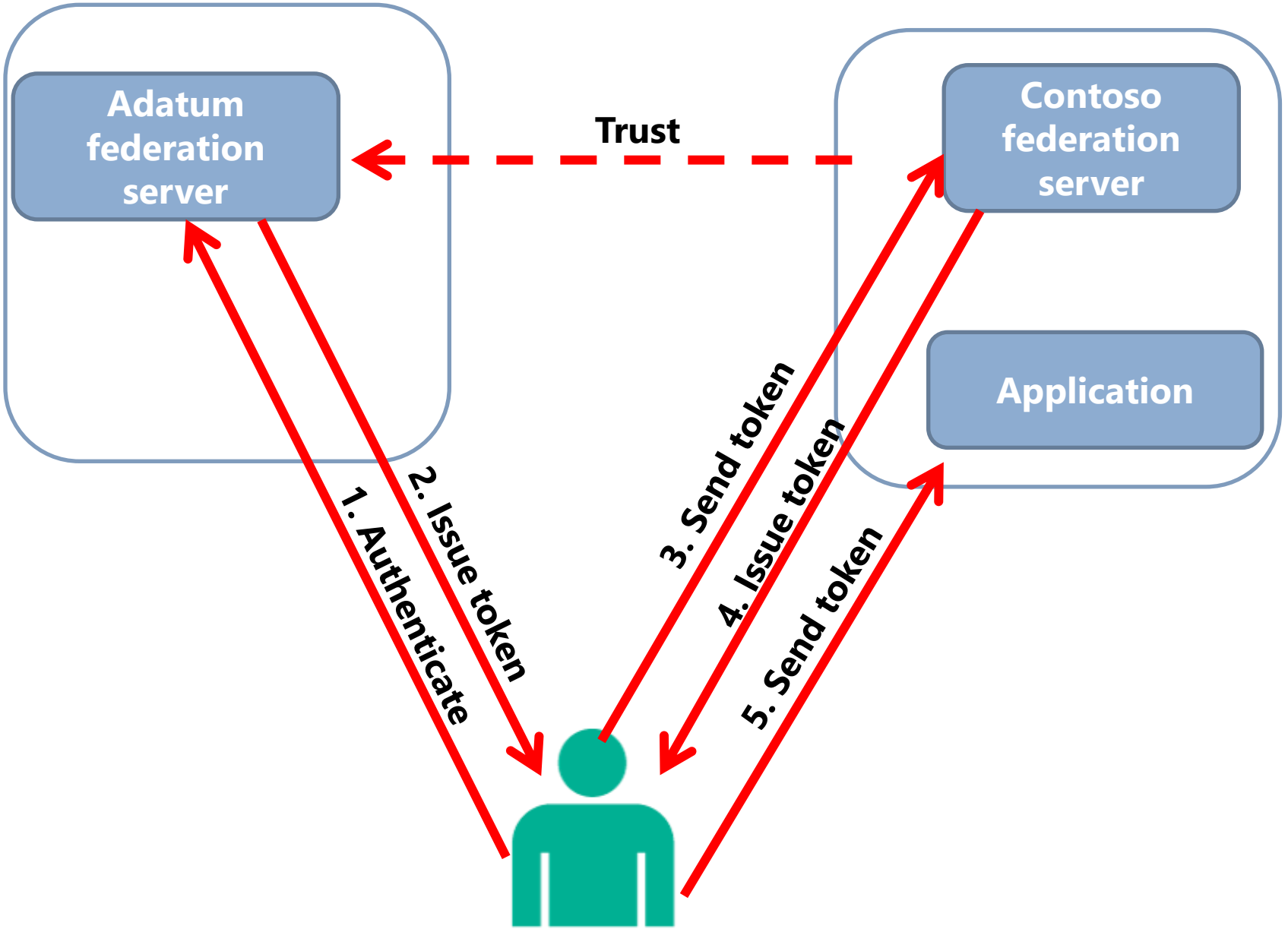
Lesson 1: Understanding identity federation

- Claims-based authentication
- Federated trusts
- What is AD FS?
- How AD FS provides SSO for Office 365

Claims-based authentication



Federated trusts



What is AD FS?

- A service provider is a relying party and is a partner in federation that creates security tokens for users
- IdP is a claims provider and is partner in a federation that consumes security tokens to provide access to applications
- The primary AD FS authentication methods are:
  - Forms authentication
  - Windows Integrated authentication
- Attribute stores are the directories or databases used to store user accounts and associated attribute values

How AD FS provides SSO for Office 365

- Windows Azure AD currently supports the following STS infrastructure to provide SSO in Office 365:
  - AD FS
  - Shibboleth IdP
  - SAML 2.0 IdP
  - Other third party IdPs
- Password synchronization provides a Same Sign-On experience
- AD FS provides a SSO experience

Lesson 2: Planning an AD FS deployment

- AD FS server roles
- Planning an AD FS deployment for Office 365
- Planning a highly available AD FS deployment
- Capacity planning
- AD FS requirements
- Configuring SSO with Microsoft Azure virtual machines

AD FS server roles

The AD FS server roles or components that provide an AD FS federated identity management solution include:

- Federation server
- Federation server farm
- Federation proxy
- Configuration database

Planning an AD FS deployment for Office 365

When planning your AD FS environment for integration with Office 365, consider the following design options:

- Remediation of AD DS
- Choice of the configuration database
- Use of federation proxies
- Configuration of Extended Protection for Authentication
- Virtualization of your AD FS infrastructure
- Server placement

Planning a highly available AD FS deployment

When planning the availability of your AD FS environment for federated authentication, you should consider the following categories:

- Federation server farm
- NLB
- Configuration database

Capacity planning

Use the following when planning for capacity of your federation servers:

- Capacity planning sizing worksheet requirements:
  - Percentage of total users expected to send authentication requests to AD FS during peak usage periods
  - Length of time the peak usage period is expected to last
  - Total number of users that will require SSO access
- Estimation table

Number of Users	Minimum Number of Servers
Fewer than 1,000	2 federation servers, 2 proxies *
1,000 – 15,000	2 federation servers, 2 proxies
15,000 – 60,000	3-5 federation servers; 2 proxies
More than 60,000	5+ federation servers, 3+ proxies

AD FS requirements

Planning for AD FS requirements includes:

- Certificates
- Hardware
- Software
- AD DS
- Configuration databases
- Browsers
- Extranets
- Networks
- Attributes
- Authentication
- Applications
- Permissions

Configuring SSO with Microsoft Azure virtual machines

Deployment scenarios for Office 365 federation components on Microsoft Azure virtual machines:

- All Office 365 SSO integration components deployed on-premises
- All Office 365 SSO integration components deployed in Azure
- Some Office 365 SSO integration components deployed in Azure for disaster recovery

Lesson 3: Deploy AD FS for identity federation with Office 365

- Installing and configuring AD FS
- Installing and configuring AD FS proxy
- Installing and configuring Web Application Proxy for AD FS
- Configuring AD FS by using Azure AD Connect
- Configuring AD FS for federation with Office 365
- Converting the Office365 tenant to federated authentication
- Managing an AD FS deployment
- Verifying SSO
- Temporary fall back to password synchronization

Installing and configuring AD FS

- The following items may need to be in prepared before installing AD FS:
  - SQL Server
  - Service account
  - Certificates
  - DNS
- During deployment of AD FS, you will:
  1. Install AD FS
  2. Configure AD FS
  3. Create the first federation server in a farm
  4. Add a federation server to a farm
  5. Update AD FS

Installing and configuring AD FS proxy

- You might need to prepare the following items before you install AD FS proxy:
  - Certificates
  - Load balancing
  - DNS
- During deployment of AD FS proxy, you will:
  - Install AD FS proxy
  - Configure AD FS proxy
  - Update AD FS proxy

Installing and configuring Web Application Proxy for AD FS

- The following items may need to be in prepared before installing Web Application Proxy:
  - Certificates
  - Load balancing
  - DNS
- During deployment of Web Application Proxy, you will:
  - Install Web Application Proxy
  - Configure Web Application Proxy
  - Update Web Application Proxy

Configuring AD FS by using Azure AD Connect

You can use Azure AD Connect to deploy AD FS in the following scenarios:

- Create a new AD FS farm or use an existing AD FS farm
- Deploy one or more AD FS federation servers
- Deploy one or more WAP servers
- Configure the AD FS service account
- Configure the federated Azure AD domain

Configuring AD FS for federation with Office 365

- After deploying the AD FS servers, you need to:
  - Set up DNS records for the AD FS federation service name
  - Configure your load balancer if deploying more than one AD FS federation server or proxy
  - Ensure that the federation service name is added to the intranet zone in Internet Explorer for each client
- Consider additional authentication mechanisms:
  - Enable device authentication with Device Registration Service
  - Mandate the user of more secure authentication methods
  - Set MFA requirements

Converting the Office365 tenant to federated authentication

- When configuring AD FS for federation with Office 365, you will need to:
  - Install Azure AD Module for Windows PowerShell
  - Deploy a federated trust between Azure AD and AD FS
- When deploying a federated trust between Azure AD and AD FS, you will:
  - Add a new SSO domain using:

    New-MsolFederatedDomain
  - Convert an existing domain to an SSO domain using:

    Convert-MsolDomainToFederated

Managing an AD FS deployment

After installation, you may need to perform periodic AD FS management tasks, including:

- Managing the certificate life cycle
- AD FS supports automatic certificate roll over which renews AD FS certificates once a year.
- Using **Get-ADFSCertificate** cmdlet to view certificate expiration dates.
- Using the **Update-MsolFederatedDomain** cmdlet to manage certificate rollover when the AD FS token-signing certificate renews on an annual basis
- Using **Set-AdfsSyncProperties** cmdlet to change the primary/secondary AD FS federation server

Verifying SSO

After deploying SSO, you should complete the following tasks to validate the SSO functionality, and diagnose issues:

- Deploy to pilot users first
- Verify with Microsoft Remote Connectivity Analyzer website
- Verify with Microsoft Connectivity Analyzer tool
- Verify federation service

Temporary fall back to password synchronization

- You can temporarily fall back to password synchronization in case of AD FS failure or other failure that makes local AD FS and AD DS unavailable
- You should use ~~Convert-MSOLDomainToStandard~~ **Set-MSOLDomainAuthentication** cmdlet ~~with the SkipUserConversion switch~~
- Fall back to password synchronization is a manual procedure
- ~~Downtime is expected~~

Lab: Planning and configuring identity federation

- Exercise 1: Deploying Active Directory Federation Services (AD FS) and Web Application Proxy
- Exercise 2: Configuring federation with Microsoft Office 365
- Exercise 3: Verifying single sign-on (SSO)

Logon Information

Virtual machines:	**20347B-LON-DC1** **20347B-LON-DS1** **20347B-LON-WAP1** **20347B-LON-CL1**
User names:	**Adatum/Administrator** **Adatum/Holly**
Password:	**Pa55w.rd**

Estimated Time: 75 minutes

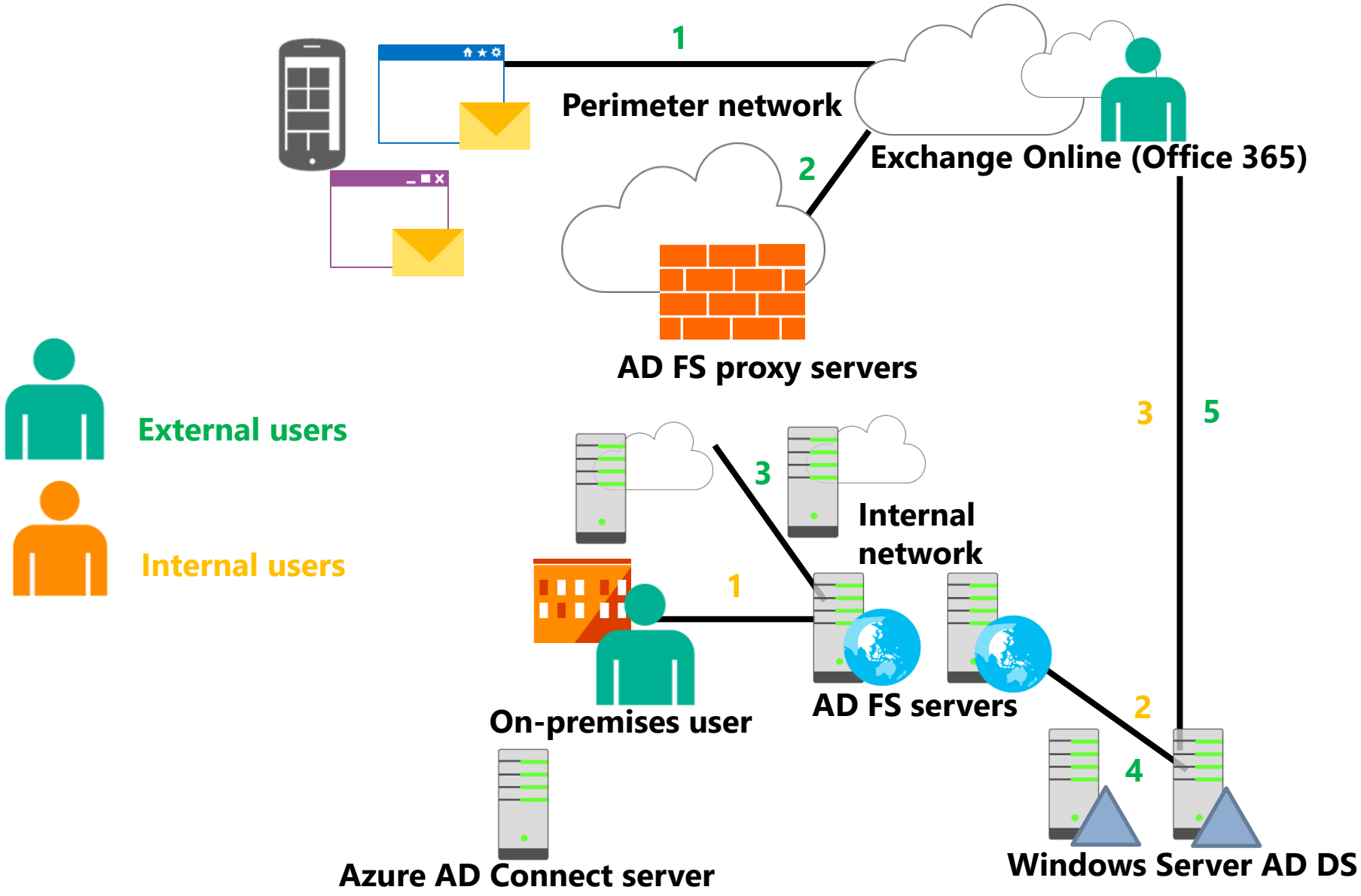
Lab Scenario

Directory synchronization is working well, and it has resolved the issue of managing user accounts in two locations. However, the security group at Adatum is concerned that users will be able to log on directly to Office 365, which reduces their options for monitoring user logons. To ensure that all users will authenticate using the on-premises AD DS domain, you have decided to implement AD FS.

Lesson 4: Planning and implementing hybrid solutions (Optional)

- Overview of Exchange Server hybrid deployment
- Configuring Exchange Server hybrid deployment
- Overview of Skype for Business hybrid
- Configuring Skype for Business hybrid deployment
- Overview of SharePoint Server hybrid deployment
- Configuring SharePoint Server deployment

Overview of Exchange Server hybrid deployment



Configuring Exchange Server hybrid deployment

Exchange Deployment Assistant

Summary: Customized step-by-step instructions to deploy Exchange Server in your on-premises organization.

The Exchange Deployment Assistant is a web-based tool that asks you a few questions about your current environment and then generates a custom step-by-step checklist that will help you deploy Exchange Server in your on-premises organization. The Deployment Assistant is currently available in English only.

Looking for help with migrating your Exchange organization to Office 365 and Exchange Online? Check out Use the Office 365 mail migration advisor in Office 365.

Launch the Deployment Assistant

Exchange 2016 scenarios

The following deployment scenarios are available for Exchange Server 2016:

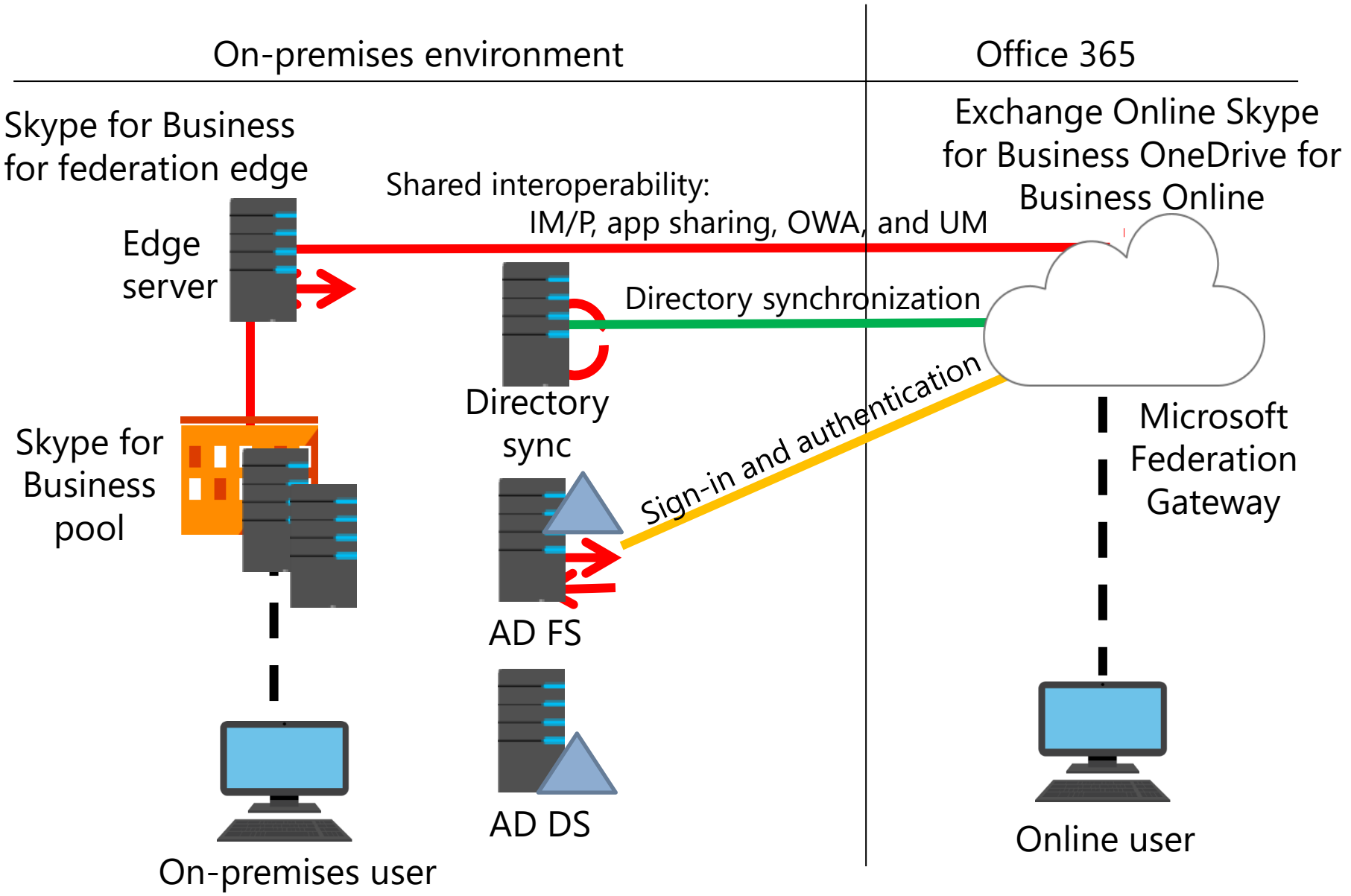
- New installation of Exchange Server 2016
- Upgrade from Exchange 2010 to Exchange 2016
- Upgrade from Exchange 2013 to Exchange 2016
- Upgrade from mixed Exchange 2010 and Exchange 2013 to Exchange 2016

Exchange 2013 scenarios

The following deployment scenarios are available for Exchange Server 2013:

- New installation of Exchange Server 2013
- Upgrade from Exchange 2010 to Exchange 2013
- Upgrade from Exchange 2007 to Exchange 2013
- Upgrade from mixed Exchange 2007 and Exchange 2010 to Exchange 2013

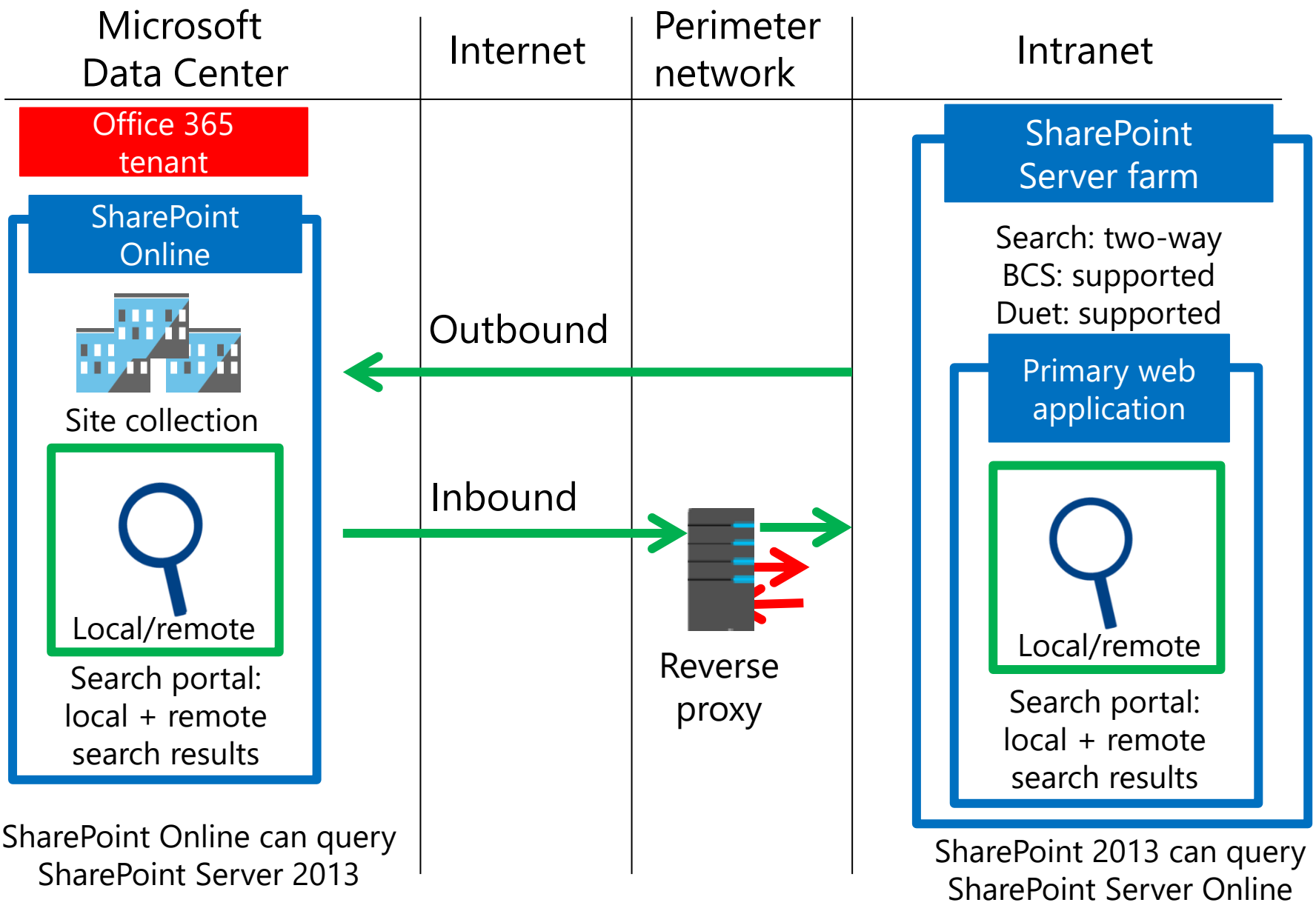
Overview of Skype for Business hybrid



Configuring Skype for Business hybrid deployment

- To deploy Skype for Business hybrid, you must:
  - Create a tenant account for Office 365
  - Add your on-premises domain to Office 365, and verify
  - Create DNS records
  - Deploy Edge Server and enable federation
  - Verify blocked/allowed domains in both environments.
- When configuring Skype for Business hybrid, you should consider the following features:
  - Directory synchronization
  - User authentication
  - Skype for Business Edge Server deployment
  - Federation
  - Client connectivity

Overview of SharePoint Server hybrid deployment



Configuring SharePoint Server deployment

- The trust relationships in SharePoint hybrid security include:
  - Identity management
  - Server-to-server trust
  - Service integration
- SharePoint Server hybrid service integration options:
  - Search
  - Sites
  - OneDrive for Business
  - BCS
  - Duet Enterprise Online

Module Review and Takeaways

- Review Question
- Real-world Issues and Scenarios
- Common Issues and Troubleshooting Tips

Course Evaluation

- Your evaluation of this course will help Microsoft understand the quality of your learning experience
- Please work with your training provider to access the course evaluation form
- Microsoft will keep your answers to this survey private and confidential and will use your responses to improve your future learning experience. Your open and honest feedback is valuable and appreciated